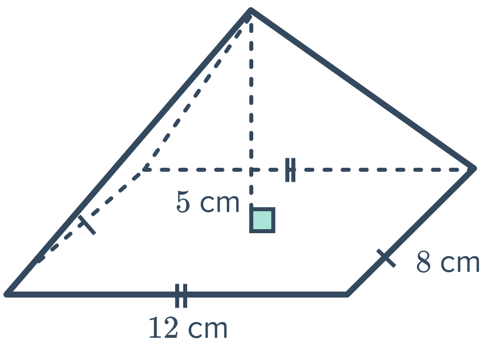
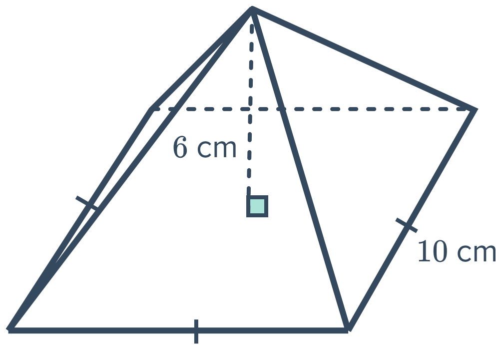
Volume of a pyramid

1 Find the volume of the following pyramids, rounding your answers to two decimal places where necessary:

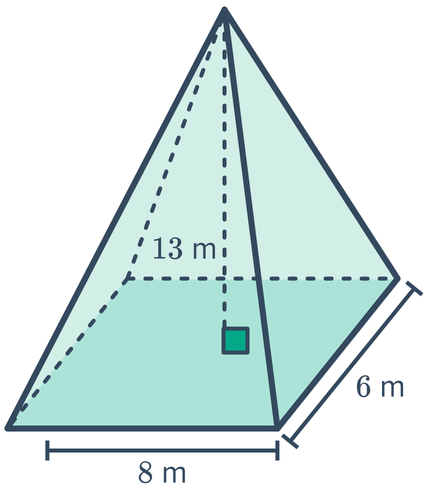
a



b



c



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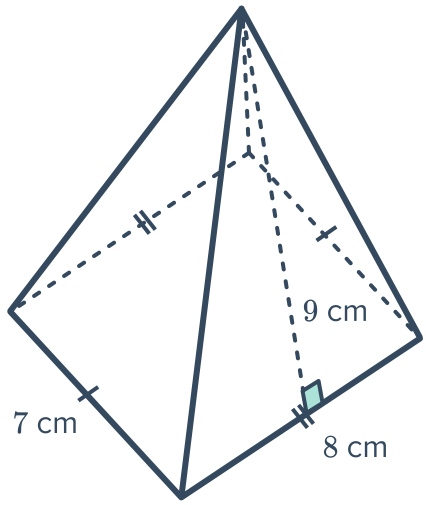
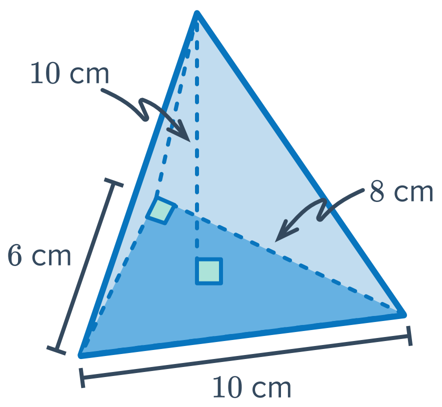
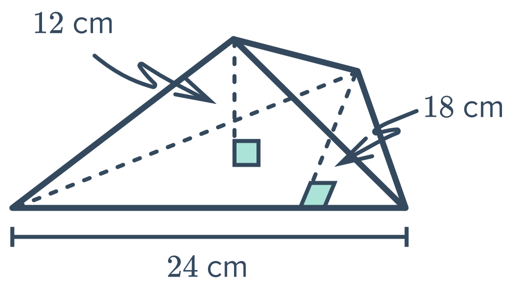


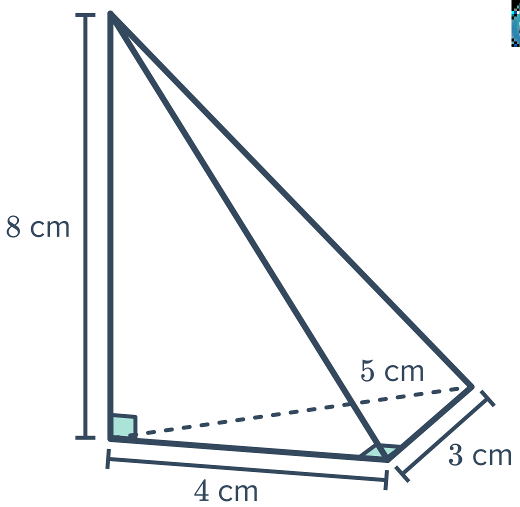
e



f



g

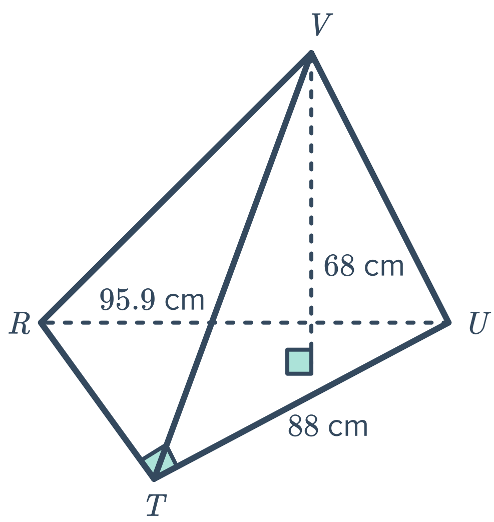


2 In the triangular pyramid shown:

a Find the length of RT in centimetres, writing your answer correct to two decimal places.

b Find the area of the base $\triangle RTU$.

c Hence, find the volume of the pyramid correct to two decimal places.



3 Find the volume of the following pyramids:

a A right pyramid with a square base of 24 m and a perpendicular height of 25 m.

b A pyramid with a height of 5 m and a right-angled triangular base with side lengths $5 \text{ m}, 12 \text{ m}$ and 13 m.

4 Find the perpendicular height of the following pyramids:

a A right square pyramid with a base of length 24 cm and a volume 2304 cm^3.

b A rectangular pyramid with a volume of 234 cm^3, a base width of 13 cm and base length of 9 cm.

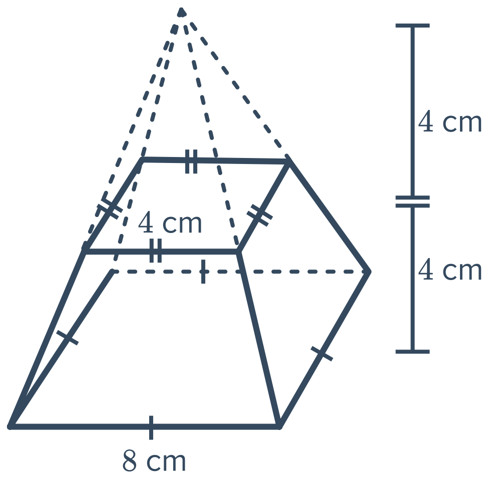
5 Find the base length of the following pyra

a A right square pyramid with a perpendicular height of 36 mm and a volume of 6912 mm^3.

b A rectangular pyramid with a volume of 180 cm^3, a perpendicular height of 12 cm and a base width of 5 cm.

6 Find the volume of a pentagonal pyramid with base area of 90 cm^2 and height 8 cm.

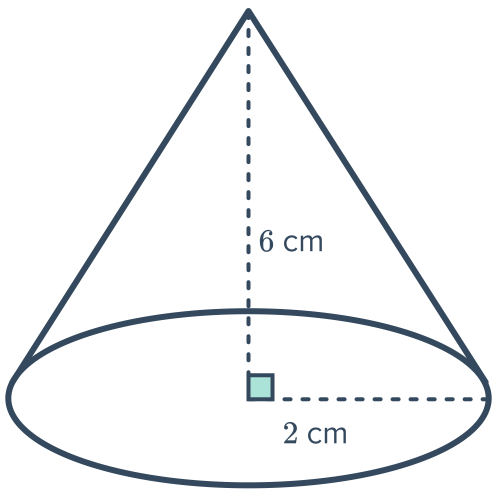
7 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown. Find the exact volume of the solid.



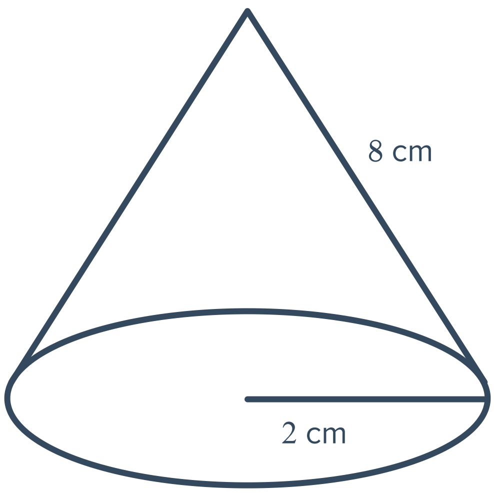
Volume of a cone

8 Find the volume of the cones shown below. Round your answer to two decimal places.

a

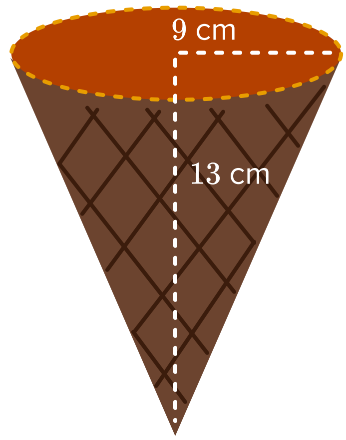


b



9 Find the capacity of the ice-cream cone in millilitres.

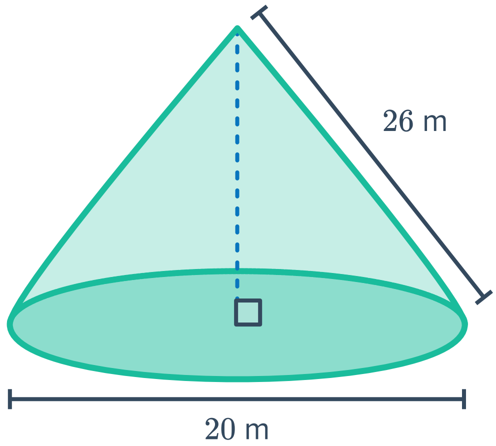
Round your answer to one decimal place.



10 Consider the cone on the right:

a Calculate the perpendicular height.

b Hence find the volume of the cone. Round your answer to three significant figures.



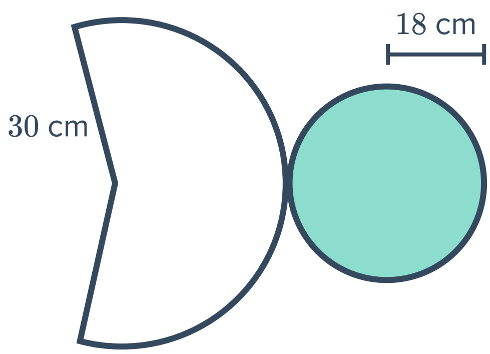
11 Consider the following net of a cone:

a Which line segment has a length of 18 cm?

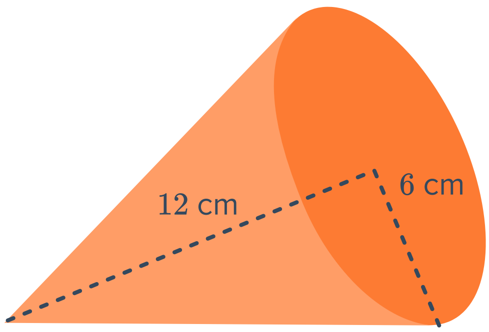
b Which line segment has a length of 30 cm?

c Find the perpendicular height, h, of the cone.

d Find the volume of the cone in cubic centimetres. Round your answer to the nearest cubic centimetre.



12 Find the volume of the cone shown. Round your answer to one decimal place.



13 Find the volume of a cone that has a radius of 8 cm and a perpendicular height 16 cm. Round your answer to two decimal places.

14 A cone has a volume of 160 mm^3. If the perpendicular height and radius of the cone are equal in length, find the radius correct to two decimal places.

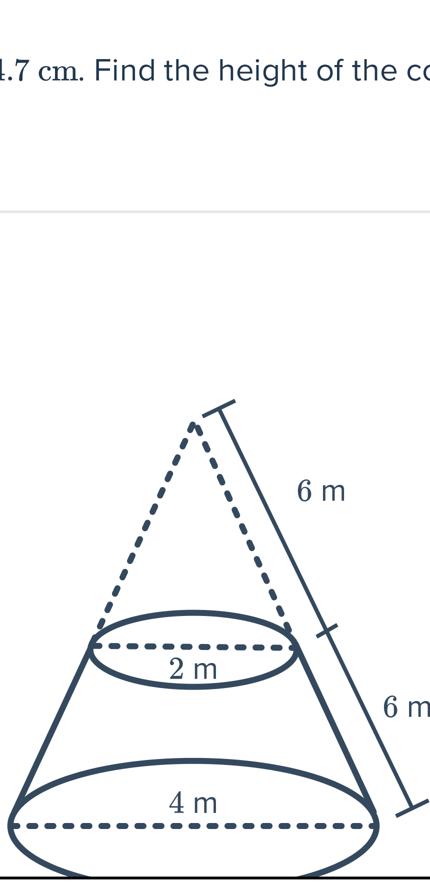
15 Find the radius of a cone that has a volume of 2456.97 cm^3 and a perpendicular height of 23 cm. Round your answer to one decimal place.

16 A cone has a volume of 1696.46 mm^3 and a perpendicular height of 20 mm. Find the radius of the cone. Round your answer to the nearest mm.

17 A cone has a volume of $106\,532.52 \text{ cm}^3$ and a radius of 54.7 cm. Find the height of the cone. Round your answer to the nearest cm.

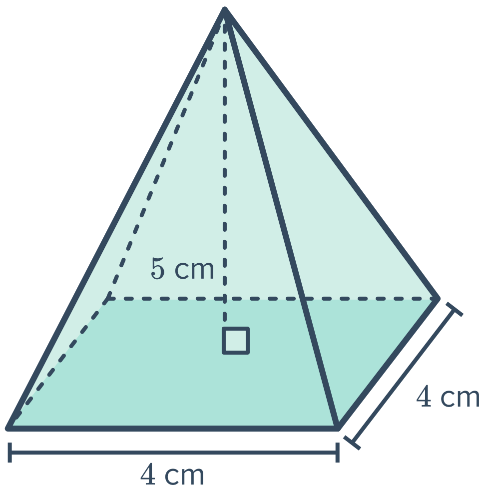
Applications

18 The following podium was formed by sawing off the top of a cone. Find its volume, correct to two decimal places.



19 A paperweight is in the shape of a square pyramid with dimensions as shown. The paperweight is filled with solid glass.

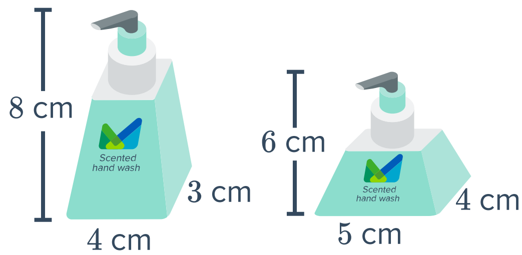
Find the volume of glass needed to make 3000 paperweights.



20 Consider the following bottles of portable hand wash. Assume that their volumes are equivalent to pyramids of the same dimensions.

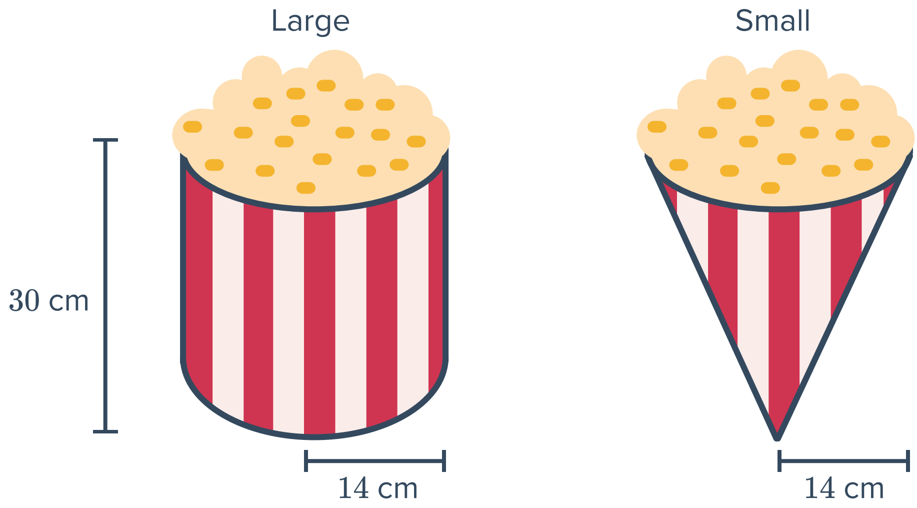
a By approximately what factor is the amount of hand wash in bottle B greater than in bottle A?

b If bottle A costs $\$50.88$ and bottle B costs $\$60.00$, which is the better buy?



Bottle A Bottle B

21 A theatre serves popcorn in two different [redacted]iners. The Small size popcorn comes in a cone-shaped container, and the Large size popcorn comes in a cylindrical container as shown below. Both containers have a radius of 14 cm and a height of 30 cm.



a How many small containers must be purchased in order to have the same amount of popcorn as in one large container?

b Find the volume of the small container. Round your answer to the nearest cubic centimetre.

c Find the volume of the large container. Round your answer to the nearest cubic centimetre.

d The cost of a small container of popcorn is $\$1.30$ and the cost of a large container of popcorn is $\$3.50$. Which is the better buy?